

1. In the circuit shown in Figure 1, find the minimum input voltage V_s required to turn on the diode with a cut in voltage of 0.7 V.

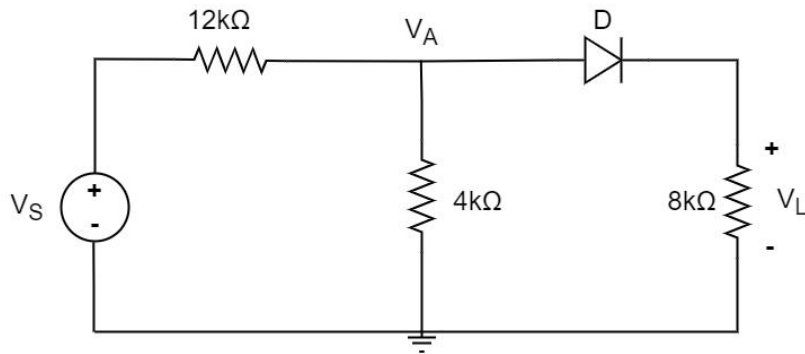


Figure 1:

2. What will be the relationship between V_o and V_i for the circuit in Figure 2

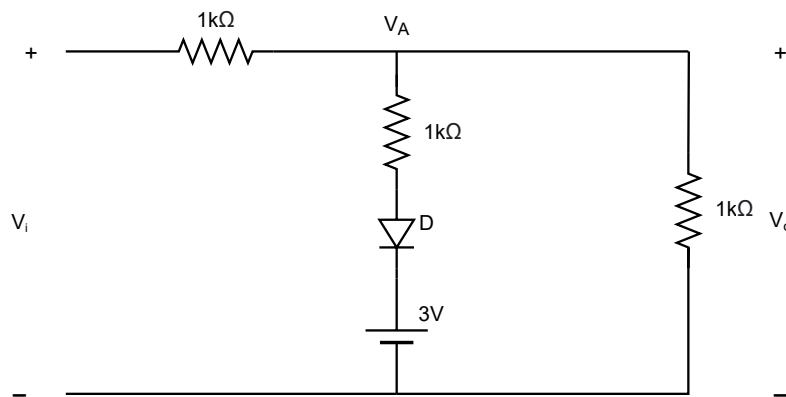


Figure 2:

3. Find $v(t)$ and $i(t)$ in the circuit shown in Fig. 3.

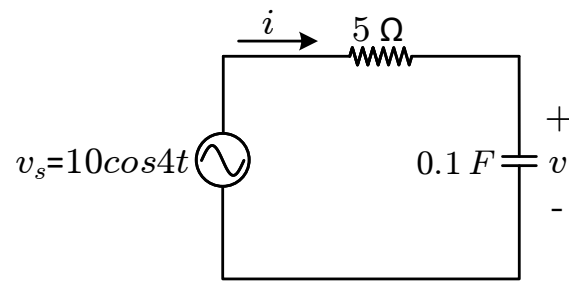


Figure 3:

4. Find the circuit's input impedance in Fig. 4. Assume that the circuit operates at $\omega = 50 \text{ rad/s}$.

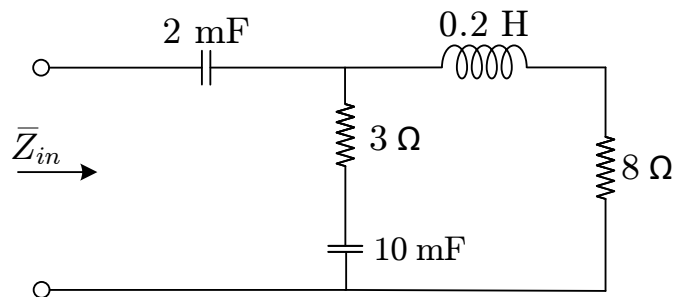


Figure 4:

5. Determine $v_o(t)$ in the circuit of Fig 5.

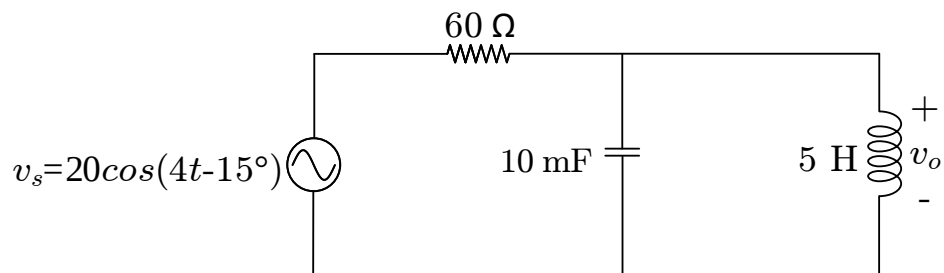


Figure 5: